

Where does this all leave us in regard to using the Kelly criterion for trade sizing? It depends. If you believe you can roughly estimate your probability of winning and the average win/average loss ratio, then the Kelly criterion can be very useful in defining the appropriate trade size. Even in this instance, however, half Kelly will be a better choice than full Kelly for most people. If, however, the probability of winning is subject to wide error, then even half Kelly is probably much too large, and the Kelly criterion would be of limited use.

¹The estimated number of atoms in the earth is 10^{50} (source: www.wolframalpha.com).

²The empirical evidence provided by track records such as Thorp's is only one of many refutations of the efficient market hypothesis. For a detailed discussion of this subject, see the forthcoming book by the author, tentatively titled *How Markets Really Work (and How They Don't)*.

³Random House, New York, 1967.

⁴The fraud investigator who repeatedly provided evidence to the SEC that Madoff was a fraud, evidence that they failed to act on.

⁵Thorp recalls the specifics of the stock parking charge as follows:

I was told that a trader at Drexel (Bruce Newberg, one of those charged in the PNP case) had a \$25 million capital line from Drexel, which he fully used. But there were more opportunities so, to exploit them, he sold some of his positions to PNP with a guaranteed buyback "up" 20 percent annualized. Using the money so obtained, he could do more good trades. The problem from the regulatory standpoint is that the parking concealed who really owned the positions.

⁶The basic strategy is a set of rules of when to stand, hit, split, and double, depending on the player's card and the dealer's card. For example, the basic strategy indicates standing with 16 if the dealer's card is between two and six, and hitting otherwise. The basic strategy does not involve any card counting.

⁷The opening paragraph in the Wikipedia entry for Claude Shannon provides the following synopsis:

Claude Elwood Shannon (April 30, 1916–February 24, 2001) was an American mathematician, electronics engineer, and cryptographer known as "the father of information theory." Shannon is famous for having